Numeric Expected Experiment Matrix v4

EXPECTED — NOT OBSERVED

Decision. This is the sole numerical expectation reference for the experiment matrix in `artifacts/aamas2027/main.pdf`. It is a falsifiable planning benchmark, not a result. The current PDF prints four RQ rows; its stated confirmation-card expansion is represented here by exactly nine cells (RQ1-I through RQ4-II). Real observations replace expectations and are never edited to match them. **Numeric basis.** The historical numeric mother is expected v3; its broken PDF is not reused. Guide v2 supplies the joint/paired uncertainty logic; guide v5 supplies positive and negative regime guards. Public work informs metric discipline and conservative scale only; no external result is treated as protocol-equivalent to ArtifactRole or PeerSelect. **Shared design.** Three structural roots × six independently reset streams per arm ($n = 18$). ArtifactRole uses 120 ITT attempts/stream and a frozen 96-case H1 cohort/stream. PeerSelect uses 1,000 warm-up plus 5,000 scored arrivals/stream. $U = Y - 0.10C$; C is complete per-attempt cost normalized to matched uniform = 1.00. Assignment change is diagnostic only.

1. Current-PDF correspondence and expected direction

PDF row	Track/unit	Current-PDF contrast	Expanded cells	Expected boundary
RQ1	ArtifactRole/root	RARE/raw/terminal/contextual trust	RQ1-I, RQ1-II	Small positive signal only under informative typed evidence; same-information strong controls remain close.
RQ2	ArtifactRole/target	Evidence+delay/no evidence/delay-only/pooled	RQ2-I, II, III	Positive under stable crossover; pooled wins in sparse support; change rate never defines success.
RQ3	PeerSelect/stream	Updater/uniform/no-history/history-only	RQ3-I, II	Payoff may improve normally; RARE loses service and utility under high load; correction buys stability at cost.
RQ4	Both/mutation+drift	Ownership/correction-replay/leave-one-out	RQ4-I, II	Safety requires risk plus coverage/UNKNOWN; stale reset may beat RARE on quality/recovery.

2. RQ1—Information value

RQ1-I: development-root signal controls. Producer-contract Q_p and independent later-use Y use separate frozen common cohorts. Brier, ECE, and log loss are lower-better; AUC is higher-better.

Target	Arm	Brier	Log loss (nats)	ECE	AUC
Producer-contract Q_p	Raw acceptance	0.225	0.637	0.073	0.680
Producer-contract Q_p	Terminal only	0.220	0.625	0.068	0.695
Producer-contract Q_p	RARE	0.203	0.584	0.055	0.738

Target	Arm	Brier	Log loss (nats)	ECE	AUC
Independent later-use Y	Raw acceptance	0.232	0.651	0.078	0.665
Independent later-use Y	Terminal only	0.218	0.620	0.066	0.704
Independent later-use Y	RARE	0.209	0.596	0.059	0.728

Expected eligible-judgment weighted $\kappa = 0.780$ [0.740, 0.820]. This is a judgment-contract diagnostic, not a policy win. **RQ1-II: confirmation-root matched-information controls, counterfactuals, and falsifiers.**

Target	Arm	Brier	Log loss	ECE	AUC	Cross-root drop	Role/status
Producer-contract Q_p	Trace only	0.216	0.616	0.064	0.704	0.019	required counterfactual
Producer-contract Q_p	Count only	0.213	0.609	0.062	0.712	0.018	required counterfactual
Producer-contract Q_p	J masked	0.212	0.606	0.061	0.716	0.018	required counterfactual
Producer-contract Q_p	J permutation	0.213	0.610	0.063	0.710	0.019	required falsifier
Producer-contract Q_p	Menu/ID permutation	0.214	0.612	0.064	0.708	0.020	required falsifier
Producer-contract Q_p	Contextual-trust-linear	0.211	0.604	0.060	0.720	0.020	qualify/freeze before run
Producer-contract Q_p	Task-conditioned ridge	0.207	0.595	0.058	0.728	0.016	expected target; qualify/run
Producer-contract Q_p	Matched composition	0.206	0.592	0.057	0.731	0.015	expected target; qualify/run
Producer-contract Q_p	RARE	0.203	0.584	0.055	0.738	0.014	qualify/freeze before run
Closest published adapter	NO-GO: no faithful protocol-preserving adapter is currently qualified						no numeric cell

Target	Arm	Brier	Log loss	ECE	AUC	Cross-root drop	Role/status
Later-use Y	Trace only	0.223	0.630	0.068	0.694	0.023	required counterfactual
Later-use Y	Count only	0.219	0.622	0.066	0.702	0.022	required counterfactual
Later-use Y	J masked	0.218	0.619	0.065	0.706	0.021	required counterfactual
Later-use Y	J permutation	0.220	0.624	0.067	0.700	0.023	required falsifier
Later-use Y	Menu/ID permutation	0.221	0.626	0.068	0.698	0.024	required falsifier
Later-use Y	Contextual-trust-linear	0.217	0.617	0.064	0.710	0.023	qualify/freeze before run
Later-use Y	Task-conditioned ridge	0.214	0.608	0.062	0.717	0.019	expected target; qualify/run
Later-use Y	Matched composition	0.213	0.606	0.061	0.720	0.018	expected target; qualify/run
Later-use Y	RARE	0.209	0.596	0.059	0.728	0.017	qualify/freeze before run
Closest published adapter	NO-GO: no faithful protocol-preserving adapter is currently qualified						no numeric cell

On Q_p, the RARE Brier edge over matched composition is -0.003; on later-use Y it is -0.004. Both targets remain close enough for a tie or strong-control win to be an allowed outcome.

3. RQ2—Assignment causality and quality–cost utility

RQ2-I: public evidence before sealed target execution. Y is future independent quality; rework is the fraction requiring rework. Planning half-widths: Y 0.012, C 0.030, U 0.011, rates 0.015.

Arm	Y	C	U	Assignment change	Adoption	Rework
No evidence/no update	0.532	1.02	0.430	0.04	0.48	0.30
Public evidence only	0.566	1.18	0.448	0.12	0.54	0.24

RQ2-II: matched 2×2 public-evidence $\times$ delayed-update factorial.

Evidence	Delayed update	Y	C	U	Change (diagnostic)
Masked	Off	0.532	1.02	0.430	0.04
Visible	Off	0.566	1.18	0.448	0.12
Masked	On	0.570	1.15	0.455	0.13
Visible	On	0.608	1.34	0.474	0.20
Interaction I		0.004	0.030	0.001	N/A

Arithmetic is exact at displayed precision: $I_U = 0.001 = I_Y - 0.10 I_C = 0.004 - 0.10(0.030)$. The near-additive interaction prevents a universal synergy claim. **RQ2-III: local personalization boundary.**

Regime	Arm	Y	C	U	Change	Max share	Min propensity
Stable crossover	Local uniform	0.548	1.00	0.448	0.08	0.34	0.05
Stable crossover	Nonlocal/global	0.592	1.25	0.467	0.16	0.32	0.06
Stable crossover	Same-cost centralized	0.602	1.34	0.468	0.18	0.29	0.07
Stable crossover	Pooled	0.600	1.28	0.472	0.14	0.31	0.06
Stable crossover	RARE	0.608	1.34	0.474	0.20	0.27	0.08
Sparse support	Pooled	0.578	1.28	0.450	0.12	0.33	0.05
Sparse support	RARE	0.561	1.34	0.427	0.11	0.36	0.04

Under stable crossover RARE is only +0.002 vs pooled and +0.006 vs same-cost centralized, while local/nonlocal diagnostics remain explicit. Under sparse support RARE loses pooled utility by 0.023.

4. RQ3—Online behavior, load, correction, and drift

RQ3-I: selected-only end-to-end service. All p50/p95 fields are milliseconds/event from event publication through sealed read/assignment and validated update; service lag is end-to-end wall time including validation, state write, queue/retry, and no-op handling. N/A means no full update path.

Regime	Arm	Payoff	C	U	Regret	Backlog p95	State KiB	Drop/timeout
Normal	Uniform	0.500	1.00	0.400	0.150	0.0	8.0	0.001
Normal	No history	0.532	1.02	0.430	0.118	0.0	8.0	0.001
Normal	History only	0.565	1.16	0.449	0.085	0.4	20.0	0.004
Normal	Matched RLS	0.600	1.31	0.469	0.050	0.8	32.0	0.005
Normal	Matched online SGD	0.598	1.29	0.469	0.052	0.7	30.0	0.005
Normal	Matched periodic refit	0.602	1.35	0.467	0.048	1.5	64.0	0.008
Normal	RARE	0.606	1.34	0.472	0.044	1.0	36.0	0.006
High load	Uniform	0.500	1.15	0.385	0.150	0.3	8.0	0.020
High load	RARE	0.582	2.25	0.357	0.068	8.0	36.0	0.110

Regime	Arm	Publish p50	Publish p95	Read p50	Read p95	Update p50	Update p95	Service lag p95
Normal	Uniform	6.0	10.0	7.2	12.0	N/A	N/A	34.0
Normal	No history	6.6	11.0	8.4	14.0	N/A	N/A	38.0
Normal	History only	9.0	15.0	14.4	24.0	19.2	32.0	76.0
Normal	Matched RLS	10.8	18.0	25.2	42.0	39.6	66.0	142.0
Normal	Matched online SGD	10.8	18.0	24.0	40.0	38.4	64.0	138.0
Normal	Matched periodic refit	12.0	20.0	20.4	34.0	72.0	120.0	185.0
Normal	RARE	13.2	22.0	27.6	46.0	40.8	68.0	155.0
High load	Uniform	18.0	30.0	22.8	38.0	N/A	N/A	110.0
High load	RARE	72.0	120.0	156.0	260.0	288.0	480.0	920.0

Regime	Arm	CPU ms	GPU ms	RAM MiB	Model calls	Input tok.	Output tok.	Tool calls	Replay ms
Normal	Uniform	6.0	0.0	64.0	0	0	0	0	0.0
Normal	No history	7.0	0.0	64.0	0	0	0	0	0.0
Normal	History only	16.0	0.0	96.0	0	0	0	0	15.0
Normal	Matched RLS	29.0	0.0	128.0	0	0	0	0	24.0
Normal	Matched online SGD	27.0	0.0	120.0	0	0	0	0	22.0
Normal	Matched periodic refit	44.0	0.0	256.0	0	0	0	0	60.0
Normal	RARE	34.0	0.0	160.0	0	0	0	0	30.0
High load	Uniform	18.0	0.0	64.0	0	0	0	0	0.0
High load	RARE	210.0	0.0	192.0	0	0	0	0	180.0

Normal RARE has only a 0.003 utility edge over the best matched updater; periodic refit is slower. Under high load RARE loses utility to uniform by 0.028 and has worse end-to-end lag, backlog, state, replay, and drop rate. PeerSelect false producer attribution is structurally N/A. **RQ3-II: correction/replay, gradual drift, and stale-version boundaries.**

Condition	Payoff	C	U	Forgetting	Recovery updates	Update p95	Backlog p95	Replay equality
Delayed: correction+replay	0.607	1.39	0.468	0.110	6.5	74.0	1.4	1.00
Delayed: no correction/replay	0.580	1.25	0.455	0.170	9.0	60.0	0.9	0.86

Gradual-drift arm	Payoff	C	U	Forgetting	Recovery	Update p95	Backlog
Frozen prior	0.566	1.02	0.464	0.230	N/A	N/A	0.0
Matched RLS	0.602	1.31	0.471	0.125	7.0	66.0	0.9
Matched SGD	0.599	1.29	0.470	0.135	7.5	64.0	0.8
Periodic refit	0.601	1.35	0.466	0.115	8.0	120.0	1.5
RARE correction/replay	0.607	1.39	0.468	0.110	6.5	74.0	1.4

Condition	Publish p95	Read p95	Update p95	Lag p95	CPU ms	RAM MiB	Replay ms	Drop
Delayed + correction	24.0	48.0	74.0	175.0	40.0	176.0	42.0	0.010
Delayed without correction	20.0	40.0	60.0	140.0	30.0	128.0	18.0	0.012
Drift frozen prior	12.0	22.0	N/A	50.0	10.0	64.0	0.0	0.005
Drift matched RLS	18.0	42.0	66.0	145.0	30.0	128.0	24.0	0.007
Drift matched SGD	18.0	40.0	64.0	140.0	28.0	120.0	22.0	0.007
Drift periodic refit	20.0	34.0	120.0	188.0	46.0	256.0	60.0	0.009
Drift RARE	24.0	48.0	74.0	175.0	40.0	176.0	42.0	0.010
Stale reset	18.0	36.0	50.0	120.0	24.0	96.0	20.0	0.006
Stale RARE	22.0	46.0	82.0	190.0	38.0	160.0	40.0	0.012

Stale-version arm	Payoff	C	U	Brier	False attribution (all)	Recovery
Version-aware reset	0.579	1.30	0.449	0.224	0.060	6.0
RARE under mismatch	0.557	1.34	0.423	0.232	0.040	8.5

Matched RLS has slightly higher drift utility than RARE (0.471 vs 0.468), so updater superiority is not presumed. The stale reset control wins utility by 0.026 and recovery; RARE retains only the safety advantage.

5. RQ4—Attribution safety and mechanism boundaries

RQ4-I: responsibility mutations. Coverage+UNKNOWN=1. False-attribution(all) uses all owner-truth-eligible attempts; false-attribution(claimed) divides the same errors by coverage. No-op remains visible because high abstention cannot manufacture a safety win.

Mutation	Coverage	UNKNOWN	FA all	FA claimed	UNKNOWN precision	No-op	Owner-truth recall
Producer-only	0.92	0.08	0.010	0.011	0.85	0.07	0.97
Recipient-only	0.06	0.94	0.002	0.033	0.99	0.93	0.98
Sink-only	0.10	0.90	0.003	0.030	0.98	0.89	0.96
Mixed ownership	0.18	0.82	0.020	0.111	0.96	0.80	0.92
Unknown owner	0.04	0.96	N/A	N/A	0.99	0.95	N/A
Resource failure	0.30	0.70	0.010	0.033	0.97	0.68	0.90

Recipient-only safety is achieved mainly by abstention (coverage 0.06), so its low all-attempt error is never reported without UNKNOWN 0.94 and conditional error 0.033. **RQ4-II: leave-one-mechanism-out with matched safety denominators.**

Variant	Y	C	U	Coverage	UNKNOWN	FA all	FA claimed
Full RARE	0.608	1.34	0.474	0.08	0.92	0.0064	0.08
No evidence/no update	0.532	1.02	0.430	0.10	0.90	0.0180	0.18
Minus ownership gate	0.588	1.27	0.461	0.75	0.25	0.1800	0.24
Minus public evidence	0.570	1.15	0.455	0.25	0.75	0.0400	0.16
Minus delayed update	0.566	1.18	0.448	0.45	0.55	0.0495	0.11
Minus locality	0.600	1.30	0.470	0.18	0.82	0.0180	0.10
Minus updater	0.580	1.20	0.460	0.20	0.80	0.0200	0.10
Minus selected-only rule	0.603	1.31	0.472	0.30	0.70	0.0390	0.13
Minus correction/replay	0.597	1.29	0.468	0.28	0.72	0.0336	0.12

The ownership gate is the principal safety mechanism: removing it raises coverage to 0.75 but raises all-attempt false attribution from 0.0064 to 0.1800 and claimed-case error from 0.08 to 0.24. No safety ranking is made from a conditional rate alone.

6. Shared cost, interval, and denominator contract

Complete-cost arithmetic. Each total includes selection, execution, publish/read, matched judgment/masked call, assignment commit, communication, model/API/token/tool, verification, update, queue/retry, repair, state storage, and replay. Failed attempts retain observed cost.

Shared base	Select	Execute	Publish/read	Judgment	Assign	Communicate	Model/API/tool	Verify	Sum
Matched base	0.08	0.30	0.10	0.15	0.02	0.04	0.17	0.01	0.87

Template	Update	Queue/retry	Repair	Storage	Replay	Total C
uniform	0.000	0.056	0.024	0.030	0.020	1.00
no_update	0.000	0.070	0.030	0.030	0.020	1.02
raw	0.040	0.084	0.036	0.042	0.028	1.10
terminal	0.070	0.091	0.039	0.048	0.032	1.15
context	0.100	0.105	0.045	0.060	0.040	1.22
task_ridge	0.140	0.112	0.048	0.072	0.048	1.29
matched_composition	0.150	0.112	0.048	0.078	0.052	1.31
matched_rls	0.140	0.119	0.051	0.078	0.052	1.31
matched_sgd	0.130	0.112	0.048	0.078	0.052	1.29
matched_periodic_refit	0.160	0.126	0.054	0.084	0.056	1.35
pooled	0.130	0.112	0.048	0.072	0.048	1.28
rare	0.160	0.119	0.051	0.084	0.056	1.34
public_only	0.060	0.098	0.042	0.066	0.044	1.18
delayed_only	0.080	0.084	0.036	0.048	0.032	1.15
history_only	0.080	0.091	0.039	0.048	0.032	1.16

Result-row traceability. Every numeric result row stores a `cost_template_id`, all 13 cost fields, and expected wall seconds. The table below lists every distinct result-row total; the automatic audit recomputes each row from its embedded breakdown.

Row-specific ID	Update	Queue/retry	Repair	Storage	Replay	Total C
row_specific_C_1.00	0.045	0.033	0.013	0.023	0.016	1.00
row_specific_C_1.02	0.052	0.037	0.015	0.027	0.018	1.02
row_specific_C_1.15	0.098	0.070	0.028	0.050	0.034	1.15
row_specific_C_1.16	0.102	0.072	0.029	0.052	0.035	1.16
row_specific_C_1.18	0.108	0.077	0.031	0.056	0.037	1.18
row_specific_C_1.20	0.116	0.083	0.033	0.059	0.040	1.20
row_specific_C_1.25	0.133	0.095	0.038	0.068	0.046	1.25
row_specific_C_1.27	0.140	0.100	0.040	0.072	0.048	1.27
row_specific_C_1.28	0.143	0.102	0.041	0.074	0.049	1.28
row_specific_C_1.29	0.147	0.105	0.042	0.076	0.050	1.29
row_specific_C_1.30	0.150	0.107	0.043	0.077	0.052	1.30
row_specific_C_1.31	0.154	0.110	0.044	0.079	0.053	1.31
row_specific_C_1.34	0.165	0.117	0.047	0.085	0.056	1.34
row_specific_C_1.35	0.168	0.120	0.048	0.086	0.058	1.35
row_specific_C_1.39	0.182	0.130	0.052	0.094	0.062	1.39
row_specific_C_2.25	0.483	0.345	0.138	0.248	0.166	2.25

Hardware/runtime envelope: four logical CPU cores per stream, no online GPU, 1,024 MiB RAM cap. PeerSelect model/API/token/tool counts are expected zero and are printed in the H3 resource table; ArtifactRole charges them in the normalized ledger. Wall time is stored per row. **Expected 95% planning intervals.** These are not observed confidence intervals and not a common standard deviation. They are root-stratified paired-stream planning ranges. Nonzero latency/state use ±10% for single arms.

Half-width	Brier	Log loss	ECE	AUC	Y/payoff	C	U	Rate	Forgetting	Recovery
Single arm	0.008	0.018	0.006	0.020	0.012	0.030	0.011	0.015	0.020	1.2
Paired contrast	0.006	0.012	0.004	0.014	0.009	0.020	0.008	0.010	0.015	0.8

Precision inputs are explicit and machine-audited: $h = 1.96s/\sqrt{18}$, with the single-arm and paired stream-SD assumptions stored metric by metric in the JSON ledger. Recomputed residual is required to be below 10^{-6}. **Denominators.** H1 uses the frozen common held-out cohort. H2 uses all sealed assignments and attempts (ITT), with per-protocol secondary. H3 uses all arrivals/attempted updates including failures, drops, no-ops, timeouts, and UNKNOWN. H4 false-attribution(all) uses all owner-truth-eligible mutations; H4 coverage/UNKNOWN uses all mutation attempts.

7. Baseline eligibility and scientific boundary

Arm/endpoint	Status	Rule
Information counterfactuals/falsifiers	REQUIRED TARGETS	Trace/count/J-mask/J-permutation/menu-ID permutation all have expected values and must run on the same cohort and cost contract.
Contextual-trust-linear	REQUIRED QUALIFICATION	Strong same-information control; own history namespace, identical public representation/read cut/menu/budget/cost.
Task-conditioned ridge	REQUIRED EXPECTED TARGET	Numeric expectation is part of the sole reference; the eventual observed comparison is inadmissible until executable and qualified.
Matched composition	REQUIRED EXPECTED TARGET	Required guard against ordinary components explaining the gain; same qualification rule.
Matched RLS/SGD/periodic refit	REQUIRED EXPECTED TARGETS	Updater comparators nested under one information policy; they prevent automatic updater-superiority claims.
Closest published adapter	NO-GO	No faithful drop-in currently preserves the causal unit, timing, selected-only rule, assignment, and complete cost; therefore it has no numeric cell.
Graph-IPD	N/A as substitute	External native-cooperation reference only; cannot fill any nine-cell result without a qualified adapter.
PeerSelect false attribution	N/A	No producer owner-truth contract.
No-update full update latency	N/A	Report its no-op path separately; it cannot be ranked as full update latency.

8. Automatic consistency audit

All 36/36 automatic checks passed: utility identities, cost-component sums, factorial identities, coverage/UNKNOWN complements, false-attribution denominator transforms, positive/negative regime guards, nine-cell identity, and absence of TBD/ambiguous ± placeholders. The scientific evidence gate remains open because this artifact has zero benchmark/API/GPU runs.

9. Scale-calibration references

Public references inform metric discipline and conservative scale; none supplies a protocol-equivalent RARE effect.

- Gneiting and Raftery (2007), Strictly Proper Scoring Rules, Prediction, and Estimation. https://doi.org/10.1198/016214506000001437. Brier/log-loss interpretation; proper scoring-rule discipline
- Guo et al. (2017), On Calibration of Modern Neural Networks. https://proceedings.mlr.press/v70/guo17a.html. Calibration as a separate endpoint rather than a substitute for discrimination
- Li et al. (2010), A Contextual-Bandit Approach to Personalized News Article Recommendation. https://doi.org/10.1145/1772690.1772758. Contextual-bandit comparison framing; domain-specific lifts are not transplanted
- Dudik, Langford, and Li (2011), Doubly Robust Policy Evaluation and Learning. https://arxiv.org/abs/1103.4601. Selected-only feedback, propensities, and off-policy evaluation
- Bietti, Agarwal, and Langford (2021), A Contextual Bandit Bake-off. https://jmlr.org/papers/v22/18-863.html. Trade-offs and no universal dominance across contextual-bandit regimes
- Saito et al. (2021), Open Bandit Dataset and Pipeline. https://arxiv.org/abs/2008.07146. Logged bandit feedback and reproducible off-policy evaluation
- Chaudhry et al. (2018), Riemannian Walk for Incremental Learning. https://arxiv.org/abs/1801.10112. Forgetting is distinct from adaptation/recovery
- Geifman and El-Yaniv (2019), SelectiveNet. https://proceedings.mlr.press/v97/geifman19a.html. Coverage and selective risk must be reported jointly

Final interpretation. RARE is expected to be modestly better only where typed evidence is informative and task–peer crossover is stable. It is expected to be more expensive, slower, and larger in state; pooled/reset/uniform controls deliberately win in sparse, stale, or high-load regimes. This complete pattern—including losses—is the sole expected benchmark.